AI Browser Extension for Teen Safety

Protecting young users from harmful online content through intelligent, real-time browser-level intervention

Made with GAMMA

THE CHALLENGE

Underage Access to Harmful Content

The Problem

Minors routinely access pornographic, gambling, betting, and dark-web content through standard browsers, leading to psychological harm, addiction, financial loss, and broader social issues affecting teenagers, families, and educational institutions.

Why Current Solutions Fall Short

Traditional parental controls require manual setup, rely on passwords, and operate at the device level—making them easy to bypass. There's a critical need for intelligent, automated protection that works in real-time.

Real-Time Protection

Automated blocking without constant supervision

AI-Powered Intelligence

Computer vision-based age verification

Browser-Level Security

Works directly where access happens

Made with GAMMA

INNOVATION

How It Works & What Makes It Unique

01

Camera-Based Age Estimation

Deep learning CNN model analyzes facial features through device camera to estimate user age in real-time

02

Instant Access Control

If estimated age is below threshold or camera is disabled, restricted sites are automatically blocked with 404 page

03

Privacy-First Design

All inference performed locally on device—no data transmitted externally

04

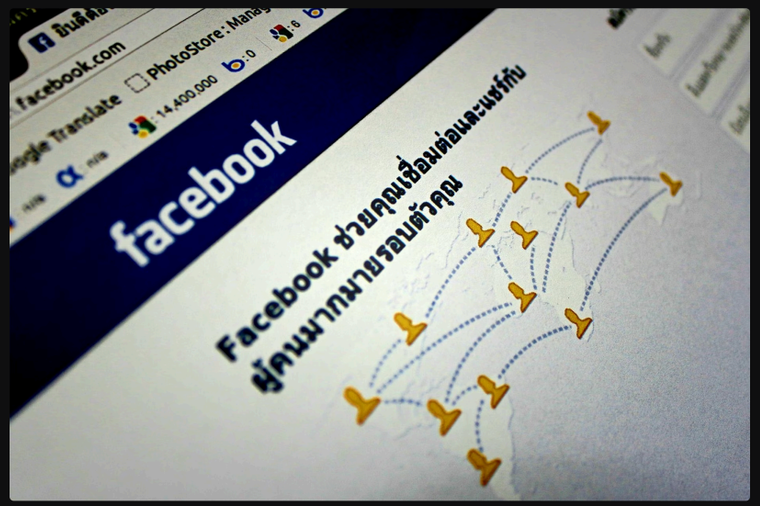
Cross-Browser Compatibility

Works seamlessly across Chrome, Firefox, Opera, and other popular browsers

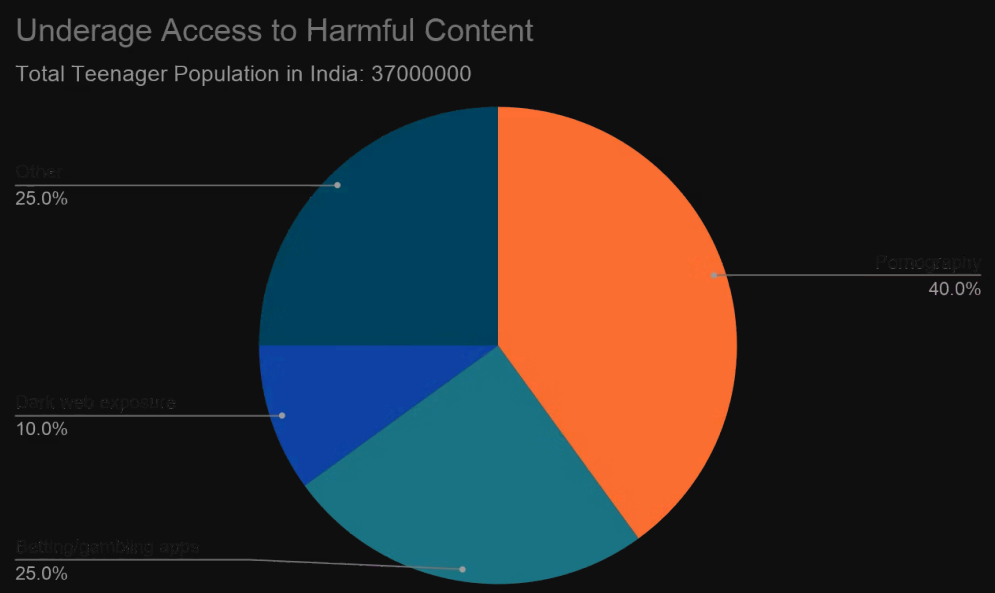
Rigorous Testing & Global Scalability

The system is trained on diverse datasets including IMDB-WIKI, UTKFace, and FairFace, evaluated through age estimation accuracy, mean absolute error (MAE), and blocking performance metrics. This first-of-its-kind browser-level biometric age gate can scale globally across schools, parental control systems, cyber cafes, and regulatory compliance frameworks.

Future enhancements: emotion detection, content severity scoring, and multilingual support to expand protection worldwide.



Made with GAMMA



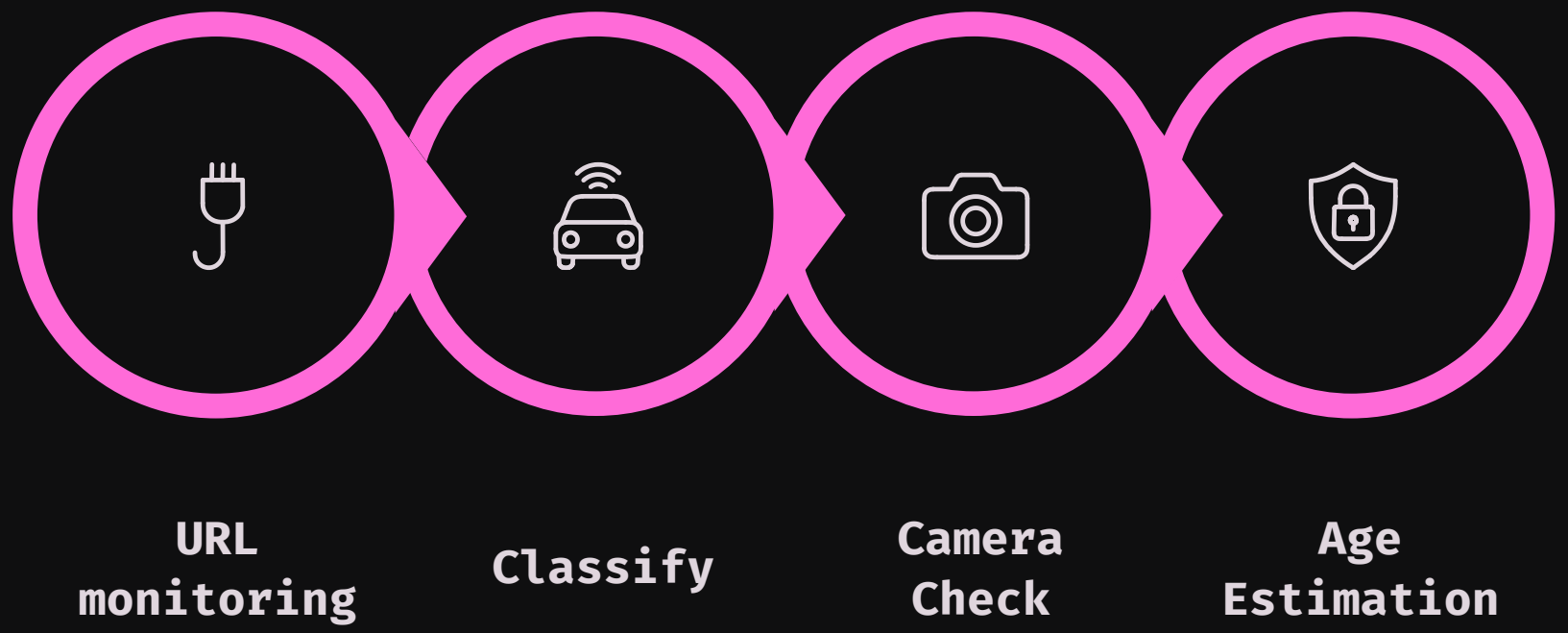
Introducing Teen Safety Shield

An intelligent browser extension designed to help teens stay protected online with real-time detection, gentle interventions, and safer browsing by default.

Made with GAMMA

System Architecture

The Teen Safety Shield functions as an integrated, multi-component system, ensuring the safety of young users online. Each element is designed to work seamlessly together, from URL monitoring to final decision-making.



This diagram illustrates the smooth path of protection, from the initial detection of content to the final access decision, ensuring a secure browsing experience.

Made with GAMMA

KEY BENEFITS

Comprehensive Protection & Peace of Mind

Our Teen Safety Shield delivers unparalleled online safety, empowering parents and safeguarding young users with advanced technology and user-centric design.

Real-time Threat Detection

Proactive blocking of harmful content through AI classification and blocklist matching, ensuring instant protection.

Accurate Age Verification

Reliable, real-time age estimation using camera-based technology, preventing underage access to restricted sites.

Unwavering Privacy

All processing occurs locally on the device, with no data transmitted or stored externally, upholding user privacy.

Broad Browser Compatibility

Seamless integration and performance across major browsers, offering extensive coverage without complex setup.

With Teen Safety Shield, families gain confidence that their teenagers are navigating the digital world securely and responsibly.

Made with GAMMA